

Math Foundations of Machine Learning

Bias and Variance

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Bias and Variance

Problem Setup

Let $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^D$ be the input features, and $y : \mathbb{R}^D \rightarrow \mathbb{R}$ be the target variable.

- The true relationship is $y_n = f(\mathbf{x}_n) + \varepsilon$, where:
 - $f(\mathbf{x}_n)$ is the **true function**
 - ε is the noise with $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{V}(\varepsilon) = \sigma^2$.
- $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ is the training dataset drawn i.i.d. from the data distribution $\mathcal{P}(\mathbf{x}, y)$.
- Let $\hat{f}(\mathbf{x}; \mathcal{D})$ be a learned model based on data $\mathcal{D}$.

We are interested in the **expected prediction error** at a point $\mathbf{x}$ averaged over all possible training sets and noise realizations.

Mean Squared Error (MSE)

The **expected squared error** of our model prediction at a point $\mathbf{x}$ is:

$$\mathbb{E}_{\mathcal{D}, \varepsilon} \left[\left(\hat{f}(\mathbf{x}_n; \mathcal{D}) - y_n \right)^2 \right] .$$

Using $y_n = f(\mathbf{x}_n) + \varepsilon$, we can rewrite this as:

$$\mathbb{E}_{\mathcal{D}, \varepsilon} \left[\left(\hat{f}(\mathbf{x}_n; \mathcal{D}) - f(\mathbf{x}_n) - \varepsilon \right)^2 \right] .$$

Expanding the square and using the independence of $\hat{f}$ and ε we get:

$$\mathbb{E}_{\mathcal{D}} \left[\left(\hat{f}(\mathbf{x}_n; \mathcal{D}) - f(\mathbf{x}_n) \right)^2 \right] + \underbrace{\mathbb{E} [\varepsilon^2]}_{\sigma^2} .$$

Bias-Variance Decomposition

Let $\hat{f}(\mathbf{x}) := \mathbb{E}_{\mathcal{D}} [\hat{f}(\mathbf{x}; \mathcal{D})]$ be the expected prediction. Then:

$$\mathbb{E}_{\mathcal{D}} \left[\left(\hat{f}(\mathbf{x}_n; \mathcal{D}) - f(\mathbf{x}_n) \right)^2 \right] = \underbrace{\left[\tilde{f}(\mathbf{x}_n) - f(\mathbf{x}_n) \right]^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}} \left[\left(\hat{f}(\mathbf{x}_n; \mathcal{D}) - \tilde{f}(\mathbf{x}_n) \right)^2 \right]}_{\text{Variance}}$$

$$\therefore \mathbb{E} \left[\left(\hat{f}(\mathbf{x}_n) - y_n \right)^2 \right] = \underbrace{\left[\mathbb{E} [\hat{f}(\mathbf{x}_n)] - f(\mathbf{x}_n) \right]^2}_{\text{Bias}^2} + \underbrace{\mathbb{E} \left[\left(\hat{f}(\mathbf{x}_n) - \mathbb{E} [\hat{f}(\mathbf{x}_n)] \right)^2 \right]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible Noise}}$$

Bias and Variance

Bias and Variance

- **Bias:** *an error introduced by very strong model assumptions.*

A very simple model (e.g., linear regression for highly nonlinear data) has high bias. As a consequence, it fails to capture the complexity of the data $\implies$ **underfitting**.

- **Variance:** *It is how much the model fluctuates with small changes in the training data.*

A very complex model (e.g., a 20th-degree polynomial) tends to memorize the training data. As a consequence, it has high variance and low generalization ability $\implies$ **overfitting**.

Bias-Variance Trade-off

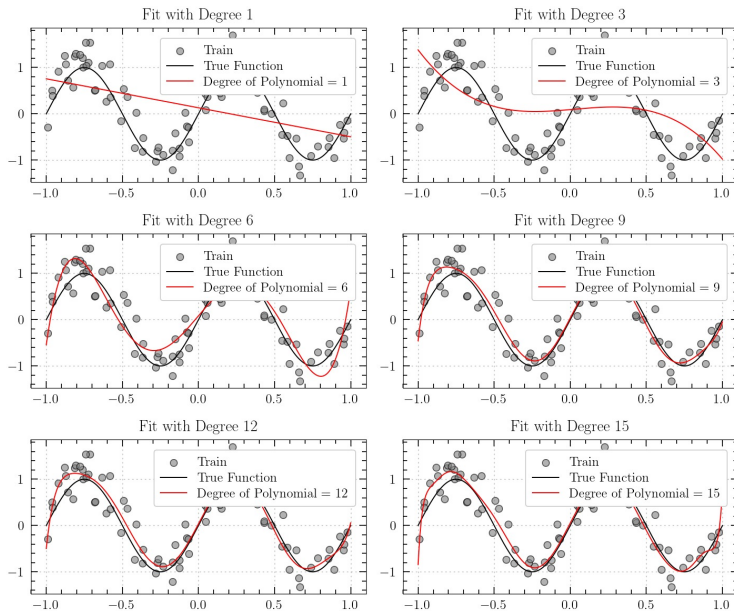
Summary

- $\uparrow$ model complexity $\implies \downarrow$ bias
- $\uparrow$ model complexity $\implies \uparrow$ variance

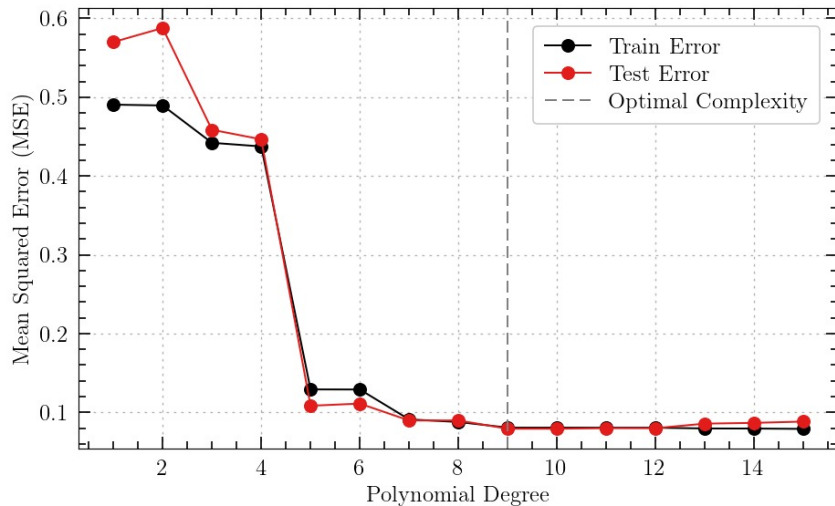
Examples

1. **Linear Regression:** High bias & Low variance $\implies$ Too simple for curved data
2. **Random Forest (very deep):** Low bias & High variance $\implies$ Can memorize noise
3. **Shallow Neural Network:** Medium bias & Medium variance $\implies$ Good balance point
4. **Very deep neural network without regularization:** Low bias & High variance $\implies$ Risk of overfitting

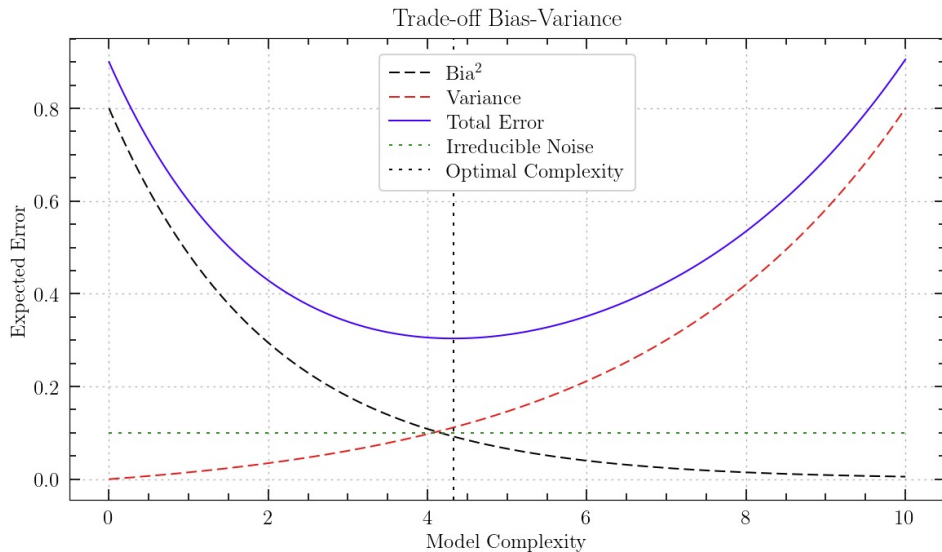
Model Complexity



Model Complexity



Model Complexity





Principal Components Analysis

Principal Components Analysis

Principal Component Analysis

- Dimensionality reduction for high-dimensional data
- Goal: Compact representation with minimal information loss
- **Applications:** compression, latent structure discovery

Intuition Behind PCA

- High-dimensional data is often redundant (overcomplete)
- Example: RGB image (640×480 921, 600D vector)
- PCA seeks a low-dimensional subspace that retains maximum information

Problem Setting

Consider an i.i.d. dataset $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subset \mathbb{R}^D$, with mean $\mathbf{0}$ that possesses the *data covariance matrix*:

$$\mathbf{S} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T.$$

Furthermore, we assume there exists a low-dimensional compressed representation (code) $\mathbf{z}_i = \mathbf{B}^T \mathbf{x}_i \in \mathbb{R}^M$ of $\mathbf{x}_i$ where we define the projection matrix:

$$\mathbf{B} := [\mathbf{b}_1, \dots, \mathbf{b}_M] \in \mathbb{R}^{D \times M}.$$

We assume that the columns of $\mathbf{B}$ are orthonormal so that $\mathbf{b}_i^T \mathbf{b}_j = 0$ if and only if $i \neq j$ and $\mathbf{b}_i^T \mathbf{b}_i = 1$. We seek an M -dimensional subspace $U \subseteq \mathbb{R}^D$, $\dim(U) = M < D$ onto which we project the data.

Problem Setting

We denote the projected data by $\tilde{\mathbf{x}}_n \in U$, and their coordinates (with respect to the basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_M$ of U) by $\mathbf{z}_n$. Our aim is to find projections $\tilde{\mathbf{x}}_n \in \mathbb{R}^D$ (or equivalently the codes $\mathbf{z}_n$ and the basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_M$) so that they are as similar to the original data $\mathbf{x}_n$ and minimize the loss due to compression.

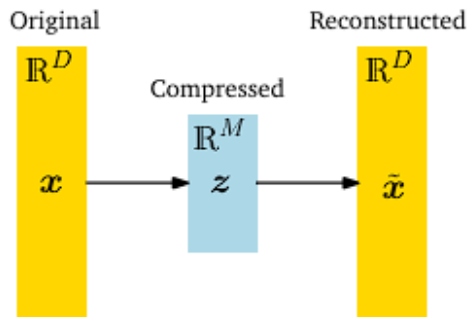
Problem Setting

1. Given data $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subset \mathbb{R}^D$.
2. Want to low-dimensional code
 $\mathbf{z}_n = \mathbf{B}^T \mathbf{x}_n \in \mathbb{R}^M$.
3. Projection matrix
 $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_M] \in \mathbb{R}^{D \times M}$.

Objective in PCA

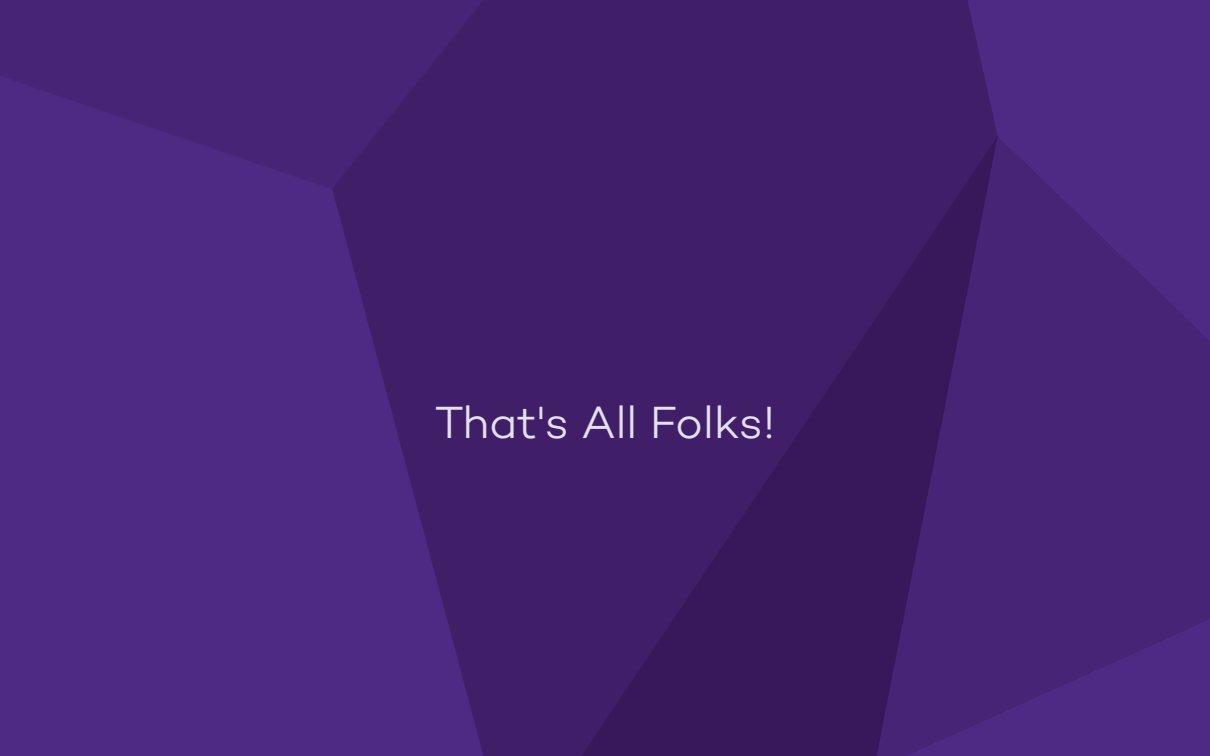
1. Find $\mathbf{z}_n$ such that $\tilde{\mathbf{x}}_n = \mathbf{B}\mathbf{B}^T \mathbf{x}_n$ is close to $\mathbf{x}_n$.
2. $\mathbf{B}$ should have orthonormal columns.
3. Compression $\implies$ fewer parameters, less storage

Problem Setting



- In PCA, we find a compressed version $\mathbf{z}$ of original data $\mathbf{x}$. The compressed data can be reconstructed into $\tilde{\mathbf{x}}$, which lives in the original data space, but has an intrinsic lower-dimensional representation than $\mathbf{x}$.

Maximum Variance Perspective

The background consists of several overlapping triangles in various shades of purple, creating a complex geometric pattern. The colors range from a deep, dark purple to a lighter, more vibrant shade.

That's All Folks!